DD-D-431D
January 3, 1977
SUPERSEDING
Int. Fed. Spec. DD-D-00431C(DSA-DM)
May 1, 1973 and
Fed. Spec. DD-D-431A
November 6, 1952

FEDERAL SPECIFICATION

DISH, BIOLOGICAL STAINING, WITH TRAY

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one type of biological staining dish with tray, suitable for use in medical treatment facilities (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Box, Shipping, Fiberboard.
PPP-B-676	- Box, Setup.

Federal Standards:

Fed. Std. No. 66	- Steel: Chemical Composition and Hardenability.
Fed. Std. No. 123	- Marking for Shipment (Civil Agencies).

FSC 6640

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers, at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-10547 - Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

American Society for Testing and Materials (ASTM) Standard:

C 162 Standard Definitions of Terms Relating to Glass and Glass Products

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Glass. The glass shall be high quality clear, soda lime, suitable in all respects for the purpose intended.

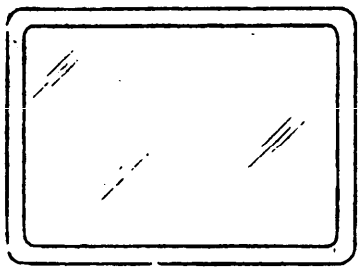
3.1.2 Metal. The metal shall be nickel copper alloy or corrosion-resisting steel conforming to AISI 302 or 304 of Fed. Std. No. 66

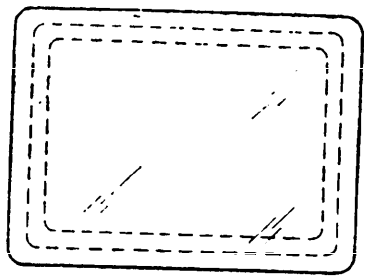
3.2 Style and design. Style and design shall conform to figure 1, and shall consist of a rectangular dish, a cover, a removable glass tray with open bottom capable of holding 20 slides, and a wire tray remover.

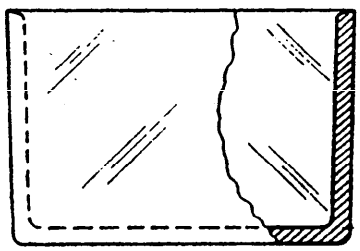
3.2.1 Dish and cover. The staining dish shall be glass, heavy-walled, and rectangular in shape. The cover shall be glass and shall have a heavy shoulder which sets inside the dish. The inside dimensions of the dish shall be: length at the top 94 ± 2 millimeters (mm.), length at the bottom 89 ± 2 mm., width 74 ± 2 mm., and height 64 ± 4 mm. The dimensions of the cover shall be such as to cause the cover to mate with the dish.

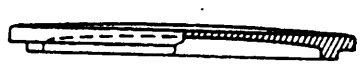
3.2.2 Tray. Tray shall be a heavy-walled, open-bottom glass tray, and shall have 10 grooves molded on the inner surface of both ends to accommodate 20 slides in pairs, back to back, measuring 73.5 to 77 mm. long by 25 to 50 mm. wide. The grooves shall be designed to enable the slides to fit snugly, bottom completely, and not interfere with the cover in the assembled dish. There shall be a hole, chamfered inside and outside, suitably located near the top of each of the end walls for insertion of the spring wire holder, so that the tray may be lifted in and out of the dish. Outside dimensions of the tray shall be: length at the top ends of the tray 89 mm., length at the bottom 84 mm., width at the top 69 mm., width at the bottom 65 mm., and height at the ends 49 mm. A tolerance of ± 2 mm. shall apply to these dimensions.

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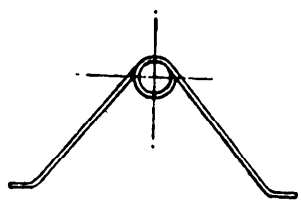






COVER

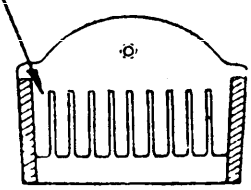
DISH



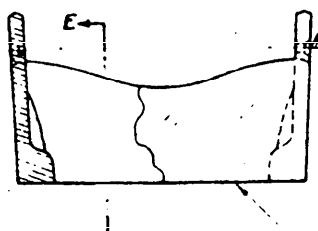
WIRE HOLDER

10 GROOVES MOULDED ON INNER SURFACE OF BOTH ENDS. EACH GROOVE SHALL ACCOMMODATE 2 SLIDES

2 HOLES FOR INSERTING WIRE HOLDER.



SECT E-E



TRAY OPEN BOTTOM

FIGURE I. STYLE AND DESIGN

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3.2.3 Wire tray remover. The wire tray remover shall be one-piece wire 0.0747 inch in diameter (nominal) (no. 13 American Wire Gauge or no. 14 U.S. Standard Gauge), suitable for lifting the tray out of the dish. A suitable design is illustrated in figure 1.

3.3 Identification marking. Each dish shall be permanently and legibly marked with the name or registered trademark of the manufacturer.

3.4 Workmanship. Workmanship shall be first class throughout. The staining dish and components shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.3 Certificates of quality. Certificate of quality, supplied by the manufacturer of the wire, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component, the lot number, the lot size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I.

TABLE I. Sampling for examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-2	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105. Inspection level shall be S-2.

4.3 Examination. The complete staining dish shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with table II.

TABLE II. Classification of defects

Categories	Defects 1/
Major:	
101	Unit not free of sharp edges.
102	Unit not free of chipping.
103	Unit not free of open blisters.
104	Unit not free of protrusions.
105	Unit not free of cracks.
106	Unit not free of checks and crizzles.
107	Unit not free of tears and deformed areas.

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TABLE II. Classification of defects (cont'd)

Categories	Defects 1/
Minor:	
201	Unit not free of stones.
202	Unit not free of seeds.
203	Unit not free of laps.
204	Unit not free of blisters, bubbles and striae,
205	Identification marking not present, complete, correct, legible and permanent.

1/ Glass defects are as defined in ASTM C 162. Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The complete staining dish shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.3.3 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing and marking requirements of section 5. Defects shall be scored as specified in table III. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with AQL of 4.0 expressed in terms of percent defective.

TABLE III. Examination of preparation for delivery

Examine	Defect
Contents	Not as specified.
Container	Not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Packing and packing materials	Component missing or damaged.
Workmanship	Bulging or distortion of container.

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4.3.3.1 Examination of closure, waterproofing and banding of container. When the container is required to comply with PPP-B-566, PPP-B-585, PPP-B-601, PPP-B-621, PPP-B-636, or PPP-B-676. The examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of the box specification.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. Each complete unit (dish with cover, tray and wire tray remover) or each dish with cover or each tray shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. The box shall contain sufficient cushioning material to prevent damage to the contents. Closure of the box shall be as specified in the appendix of the applicable box specification.

5.1.1.2 Intermediate package. Unless otherwise specified, twelve complete units (dish with cover, tray and wire tray remover) or twelve dishes with covers or twelve trays shall be packaged in a box conforming to PPP-B-636, class domestic. Closure of the box shall be as specified in the appendix of the box specification.

5.1.2 Level C. Each complete unit (dish with cover, tray and wire tray holder) or each dish with cover or each tray shall be packaged in a manner that will afford adequate protection against physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Unless otherwise specified, two intermediate packages (24 unit packages) containing complete units (dish with cover, tray and wire tray holder) or three intermediate packages (36 unit packages) containing dishes with covers or four intermediate packages (48 units packages) containing trays shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

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5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in the appendix of PPP-B-636.

5.2.2 Level B. Unless otherwise specified, two intermediate packages (24 unit packages) containing complete units (dish with cover tray and were tray holder) or three intermediate packages (36 unit packages) containing dishes with covers or four intermediate packages (48 unit packages) containing trays shall be packaged in an exterior container designed for a type 2 load and conforming to PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The packaged dish with cover or tray or the complete unit (dish with cover, tray and were tray holder) shall be packed in shipping containers that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with Fed. Std. No. 123.

5.3.2 Military activities. Each unit, intermediate package and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. This specification covers a biological staining dish with cover and removable tray and tray remover; a replacement biological staining dish with cover, and a replacement tray.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) National Stock Number (NSN).
(b) Title, number and date of this specification.
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.3 This specification covers the following items shown in the Federal Supply Catalog:

National Stock No.	Item Identification
6640-00-422-3810	DISH, BIOLOGICAL STAINING, 20 Slide
6640-00-422-3815	DISH, BIOLOGICAL STAINING, 20 Slide
6640-00-422-3820	TRAY, BIOLOGICAL STAINING Dish, Glass

6.4 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification or those which are commercially available, but is intended to cover the type normally procured to meet Federal requirements.

Military Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA-DM

Project No. 6640-1000

Civil Agency Coordinating Activities:
HEW-PHS
VA-DMS
GSA-FSS

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No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
DD-D-431D DISH, BIOLOGICAL STAINING, WITH TRAY

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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